ECE 310 Fall 2026
Midterm 1 Review

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True/False Section: FA2019, Problem 1

(a) An LSI system specified by the following difference equation: $y[n] - \frac{1}{2}y[n-1] = x[n]$ can be causal or anti-causal. **T/F**

(b) The input and output relationship of an arbitrary system is completely determined by the system's unit pulse response. **T/F**

(c) If an LSI system is BIBO unstable, its unit pulse response $h[n]$ must be unbounded. **T/F**

(d) Let $h[n] = h_1[n] * h_2[n]$ be the unit pulse response (UPR) of two serial subsystems with UPR $h_1[n]$ and $h_2[n]$. If h_1 or h_2 is BIBO *unstable*, h must be BIBO *unstable*. **T/F**

(e) An LSI system with a finite-length impulse response can be BIBO stable or unstable. **T/F**

(f) A time-varying system cannot be causal. **T/F**

(g) Let $\sum_{n=-\infty}^{\infty} x[n]\delta[2^n u[n] - 8] = 4$. Then, $x[3] = 2$. **T/F**

(h) Suppose that the step response $g[n]$ of an LSI system, i.e., the output of the system to the input $x[n] = u[n]$ has a z-transform with a pole at $z = 1/2$. Then, $H(z)$ has also pole at $z = 1/2$. **T/F**

(i) If the response $y[n]$ of an LSI system to the input $x[n] = 3^n u[n]$ is unbounded, the system must be BIBO unstable. **T/F**

(j) The response $y[n]$ of a BIBO *unstable* LSI system to any non-zero input $x[n]$ is always unbounded. **T/F**



Identifying LTIC Systems Problem: SP2023, Problem 2

	Linear	Shift-Invariant	Causal	Stable
$y[n] = x[n] * (-1)^n u[n]$				
$y[n] = \frac{x[n]}{x[2]}$				
$y[n] = \cos^2\left(\frac{\pi}{2} n\right) x[n]$				

Convolution Problem: FA2023, Problem 3

For each of the following parts, calculate the convolution $y[n] = x[n] * h[n]$.

(a) $x[n] = \{\underset{\uparrow}{1}, 2, 3, 2, 1\},\ h[n] = \{\underset{\uparrow}{-1}, 1\}$

(b) $x[n] = \left(\frac{3}{4}\right)^n u[n],\ h[n] = 2u[n] - u[n-1] - u[n-2]$

Computing z-transform Problem: FA2024, Problem 5

For each of the following discrete-time signals, find its z-transform including ROC.

(a) $x[n] = (n+1)u[n-1]$

(b) $x[n] = u[n-1]u[3-n]$

(c) $x[n] = 3^{-n}u[n] + 3^{n}u[-n]$



z-transform LTI System Response Problem: SP2023, Problem 6

Suppose the input $x[n]$ to a **causal and stable LTI system** produces output $y[n]$.

$$X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})}$$

$$Y(z) = \frac{1}{(1-\frac{1}{2}z^{-1})(1-z^{-1})(1-\frac{1}{4}z^{-1})}$$

(a) Find the transfer function $H(z)$ and its ROC.

(b) Find the impulse response $h[n]$.



z-transform LTI System Response Problem: SP2023, Problem 6

Suppose the input $x[n]$ to a **causal and stable LTI system** produces output $y[n]$.

$$X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})}$$

$$Y(z) = \frac{1}{(1-\frac{1}{2}z^{-1})(1-z^{-1})(1-\frac{1}{4}z^{-1})}$$

(c) Determine the difference equation of the system.



z-transform Stability Problem: SP2023, Problem 7

The transfer function of a causal LTI system is given below:

$$H(z) = \frac{z-3}{z-4}, \text{ ROC: } |z| > 4$$

(a) Find a bounded input $x[n]$ that will produce an unbounded output $y[n]$.

(b) Find an unbounded input $x[n]$ that will produce a bounded output $y[n]$.

(c) Find a bounded input $x[n]$ that will produce a bounded output $y[n]$.

